Written Update

August 21, 2026

This update covers the estimation side of the project: the primary specification and its timing, what the current null does and does not say, the data-quality checks completed since we last spoke, the country-level panel, and a proposed ordering of next steps. The descriptive figures are in the accompanying note, *Plots and Explanations*.

1. The estimating equation, and its timing

For firm i, holder group $h \in \{\text{US}, \text{non-US}\}$ and quarter t, the primary specification is

$$\Delta w_{iht} \;=\; \beta_2 \left(\text{US}_h \times \text{CN}_{i,t-1}\right) \;+\; \beta_3 \left(\text{US}_h \times \text{CN}_{i,t-1} \times S_{t-1}\right) \;+\; \alpha_{it} + \alpha_{ht} + \alpha_{ih} + \varepsilon_{iht},$$

estimated with the three pairwise fixed effects absorbed and standard errors two-way clustered on firm and quarter. The outcome Δw is the change in the group's portfolio weight on the firm, computed against that group's full global equity book. CN is the firm's share of supply-chain links running to China. S is the geopolitical tension shock.

Both right-hand-side terms are lagged one quarter, as agreed on 4 August: CN is measured at $t-1$ and the shock at $t-1$, with the holdings response measured at t. Specifications carrying the contemporaneous shock S_t are retained and labelled as timing diagnostics, not as the primary.

One property of the shock is worth stating because it qualifies what "lagged one quarter" delivers. S is the AR(1) innovation of the monthly bilateral tension index, and only the quarter-closing month is kept. So S_{t-1} is the March, June, September or December innovation of the previous quarter, and the other two months of that quarter are not used. The shock is also mildly serially correlated, $\text{corr}(S_t, S_{t-1}) = 0.27$, so the lagged and contemporaneous cells are not orthogonal tests. Two alternative constructions address this directly: summing the quarter's three monthly innovations, and fitting the AR(1) at quarterly frequency to begin with. Both were run at both timings and the null holds throughout (three-month sum, lagged: $p = 0.34$ to 0.58; native quarterly, lagged: $p = 0.29$).

2. How firms with no links are treated

This answers Stefano's question of 4 August. The zero-exposure group is not one thing, and that is the part worth clarifying. A firm's exposure is the share of its active customer and supplier relationships that run to China. Three cases produce something that looks like a zero.

A firm may be covered by the relationship data, have supply-chain relationships, and have none of them running to China. Its exposure is a measured zero, and it is the natural comparison for an exposed firm. A firm may be covered but carry no active supply-chain relationship of any kind, in which case the ratio has no denominator; we set it to zero rather than dropping the firm, so it enters as a zero without any measurement behind it. A firm may be absent from the relationship data altogether, and that firm is left unclassified and excluded rather than treated as a zero.

The first two are pooled in every figure and in the regression sample, and the second kind is the larger part. At the end of 2023 the split is 4,276 firms with a positive supply-chain denominator and no Chinese counterparty against 4,241 with no supply-chain links at all, of which 2,926 have no recorded relationship of any type. Pooled across quarters the no-denominator kind is about 65 percent of the zero group.

This matters for two things. Separating the two would change what the zero arm of Figure A means, since most of that arm is currently firms with nothing to measure rather than firms measured at zero. It also affects the quartile figures, which is why the no-ties line there is drawn separately rather than folded into the bottom quartile. Whether the second kind belongs in the comparison group at all is a question we would like to settle on the call.

3. What the estimand is, and what it is not

The estimation compares one treated holder group, US investors, with a single merged non-US comparison group. The coefficient of interest is therefore a group-level average: it asks whether the aggregate US position in China-linked European firms was reallocated differently from the aggregate non-US position. It answers the aggregate-reallocation question. It does not identify the government-compliance mechanism, because compliance operates at the level of individual institutions and mandates. Investor-level heterogeneity is the agreed next step for that question. The DDD structure mitigates the common firm-fundamentals alternative, but it does not by itself rule out US-specific funding, regulatory, or client-preference shocks.

4. Precision, not proof of zero

The primary coefficient is close to zero: $\beta_3 = -1.39 \times 10^{-7}$ (SE 1.03×10^{-6}, $p = 0.89$, $N = 909{,}638$, two-way clustered on firm and quarter, 10,250 firm and 82 quarter clusters). The 95% confidence interval is $[-2.17 \times 10^{-6}, +1.89 \times 10^{-6}]$ and the minimum detectable effect at 80% power and 5% size is 2.89×10^{-6}. The result is a bounded null, not evidence of a true zero.

That bound is looser than it needs to be, for a reason worth correcting. The comparison arm is defined as every non-US investor country, which puts Chinese investors themselves in the control group. They are 2.95 percent of the non-US global book that forms the denominator of the outcome, rising to 9.2–9.8 percent in the June and December quarters and falling to 0.00 percent in March and September on the semiannual reporting cycle, while contributing 0.0063 percent of the non-US holdings of European securities that form the numerator. The control arm's portfolio weights therefore inherit a large alternating movement in the denominator that carries almost no information about European firms. Removing Chinese investors from the control arm, on the same panel and the same specification, leaves the null intact and sharpens it considerably: $\beta_3 = +1.07 \times 10^{-7}$ with SE 3.59×10^{-7} ($p = 0.77$, $N = 909{,}638$), an SE roughly one third of the reported one, implying a minimum detectable effect near 1.0×10^{-6}. Adding Hong Kong to the exclusion changes nothing ($\beta_3 = +1.05 \times 10^{-7}$, SE 3.59×10^{-7}, $p = 0.77$).

We report the wider bound as the headline because the specification that produced it is the pre-registered one, and the tighter bound as the correction we believe is right. Whether Chinese investors belong in the control group is a question for the call: the literal reading of the design keeps them, since they hold China-linked firms and face no US government pressure, but their reporting cycle is what widens the interval. The same denominator hygiene issue is not unique to China, and is listed as an open item below.

5. The three-quarter horizon and the inference hierarchy

At the three-quarter cumulative horizon the wild cluster bootstrap gives $p = 0.043$. The family-wise adjusted value across horizons is 0.132. The circular-shift randomization gives $p = 0.329$. We report the circular-shift number as the headline for this cell, following the pre-registered hierarchy. The

reason is specific. The shock series is serially correlated, so the quarter-independence approximation that underlies the bootstrap becomes more suspect as the horizon lengthens. The circular-shift test preserves the shock's own autocorrelation structure under the null. This is a choice of the more conservative and better-matched reference distribution at long horizons. We do not claim the wild cluster bootstrap is invalid.

6. The fund-grain snapshot rule

The holdings data are read at fund grain with an as-of snapshot rule. Three separate reappearance-window diagnostics were run to check that rule. All three point the same way. We describe them as strong and consistent direct evidence for the fund-grain rule, not as a proof.

7. The data-quality filter, its residual, and the four-version check

The shipped filter is V1. It is a union of two legs, applied only to positions above $5bn. The market-cap leg fires when the position's market value exceeds the issuer's inline market cap. The shares leg fires when the holding exceeds shares outstanding. A row is dropped if either leg fires. V1 drops exactly 3 rows, $452.6bn in total. Each of the three fails both legs, so the leg decomposition does not change V1's dollar count.

We reran the pipeline under four filter versions. V0 drops nothing. V1 is the shipped filter, 3 rows and $452.6bn. V2 is the shares leg only, applied at any size, with no market-cap proxy anywhere in the rule; it drops 304 rows and $483.0bn. V3 is V2 plus a sentinel-price screen, 30,586 rows and $496.8bn. V2 is not V1 re-evaluated at a lower size threshold. It is narrower than V1 in one dimension and wider in another. It is narrower because it drops the market-cap leg. It is wider because it drops the size threshold. It was built that way on purpose. The market-cap proxy is derived from the same feed the filter is meant to police, so a proxy error could manufacture the filter's own justification, and V2 is the version that cannot.

Two residual figures follow from those two definitions. They count different sets of rows, and both are correct. The first is the union-leg residual below the size threshold. Running both legs with the threshold set to zero drops 376 rows and $499.6bn. Subtracting V1 leaves 373 rows and $47.0bn, spread over 196 holder-country-quarters. The second is the shares-leg-only residual. V2 minus V1 leaves 301 rows and $30.38bn; 163 of the quarter-end holder-country cells in the deltas table move under V2. The union figure is the larger of the two because it also catches sub-threshold rows that fail only the market-cap leg. Neither figure supersedes the other. Any text that reports one of them has to say which one it is.

Across the 19,340-row deltas table, 1,408 country-quarter cells move at all. Eight of those cells move by more than 1% of their own country-quarter total. No cell moves by as much as 1% of the global V1 book. The movers are the ones the filter was built around. Under V0 the dropped rows return: Poland 2017Q4 at +1479.7% and +$239.7bn, Canada 2020Q4 at +9.74%, and the United States 2019Q4 at +0.80%. Under V2 and V3 the sub-threshold residual comes out: Indonesia 2017Q4 at −65.9% and −$4.20bn, falling from $6.38bn to $2.18bn, Slovenia 2015Q2 at −4.7%, and Mauritius 2004Q4 at −1.7%.

The two remaining checks previously had a narrower scope than the deltas table. The country-ranking comparison was run at 2018Q4 and 2022Q4 only, and the Figure-A bucket-share comparison at 2018Q4, 2020Q1 and 2022Q4 only; both spot sets excluded 2017Q4 and 2020Q4, exactly the quarters carrying the two largest anomalies. Both checks have now been run on every quarter, with

per-quarter maxima, and with four quarters reported by name whatever the maxima turn out to be. The results follow.

All-quarter result, ranking. Over the 100 quarter-ends of the ranking calendar (1999Q1–2023Q4) and the three non-V1 versions, 4 country-rank positions differ from V1, in 2 of the 300 version-quarters; every one of those is a single-place swap and all four are under V3. The ordered top five is identical to V1 in all 300 version-quarters, and so is the top-five set. The largest value move for any one EU country is 5.13 percent (V3, 2008Q2, Greece), and it moves no rank.

One caveat on that denominator, which the earlier spot-quarter version of this check did not surface. The ranking family is built on US investors holding EU securities, and none of the three rows the shipped filter drops satisfies both conditions: two are US securities and the third is held by a Polish investor. V0, the do-nothing version, is therefore identical to V1 at every quarter by construction rather than by evidence, and one third of the 300 version-quarters cannot move. The evidential denominator for the ranking claim is the 200 V2 and V3 version-quarters.

All-quarter result, Figure A bucket shares. On the bucket calendar (66 quarters, 2007Q3–2023Q4; the membership file carries a 67th quarter, 2024Q1, which lies outside the holdings panel and is excluded and reported), the largest share movement against V1 is 3.6444pp on the global denominator and 8.7881pp on the EU denominator, both attained at the same cell: V0, equal-weighted measure, 2017Q4, bottom quartile, non-US holders.

That maximum belongs entirely to V0, which is not a candidate rule but the null of applying no filter at all. Among the three impossibility filters that are actually candidates, the figure barely moves: the largest movement across V2 and V3 over the full calendar is 0.021pp global (V3, value-weighted, 2017Q4) and 0.060pp on the EU denominator (V3, value-weighted, 2018Q2). The result is therefore a vindication of the shipped rule rather than a robustness problem. What the exercise does establish is that not filtering at all is not an option: the single Polish row is 3.8 percent of the non-US global book in 2017Q4.

Named quarters, reported regardless of the maxima. The two originally named quarters were identified from the holder-country deltas table and are both holder-country anomalies, 2017Q4 Polish and 2020Q4 Canadian, so the US-held ranking family cannot move on either and its zeros there are structural rather than evidential. The named set has been widened to the two quarters where the US-held book does move, 2016Q4 and 2018Q2. Across all four named quarters and all three non-V1 versions the ranking shows 0 rank moves; the largest value moves are 0.705 percent at 2016Q4 (V2 and V3) and 0.426 percent at 2018Q2 (V3). On the bucket side the two original quarters do carry the anomalies, since the Polish and Canadian holders sit in the non-US group: 2017Q4 moves 3.6444pp global and 8.7881pp EU under V0, and 2020Q4 moves 0.1533pp global and exactly 0.0000pp EU, the latter because the Canadian holder's security is American and never enters the EU book.

8. Country-panel results (Step 4)

The outcome here is the change in log dollar holdings at the country-quarter level. It is not the portfolio-weight estimand you specified on 4 August. The country-level weight panel has not been built yet, so we cannot report the object you asked for at this level; that is an open item, not a result.

The US leg is clean and is the part worth stating. Regressing the change in log US holdings on the lagged interaction of tension with country exposure gives coefficients indistinguishable from zero on all three measures: M1 $b = -0.025$ ($p = 0.77$, $N = 1{,}940$), M2 $b = -0.076$ ($p = 0.44$, $N = 2{,}016$), M3 $b = -0.008$ ($p = 0.98$, $N = 2{,}036$). This is the leg that speaks to the US

perspective, and it shows nothing.

The differenced block, US minus non-US, is negative on all three measures (M1 −0.695, M2 −0.459, M3 −1.310), but we do not think that pattern is interpretable, for three reasons we checked directly.

First, it is carried by two or three small countries. Dropping Poland from M1 moves the coefficient from −0.695 to −0.421 and the p-value from 0.02 to 0.21. Dropping Lithuania and Latvia from M2 collapses it from −0.459 to −0.167. Dropping Latvia from M3 halves it, from −1.310 to −0.655. Latvia is the clearest case: within the estimation samples its exposure measure is exactly zero in every one of its rows, so by construction it cannot respond to the interaction, yet removing it moves the estimate substantially. Effective cluster counts, adjusting for how concentrated the regressor variance is across countries, are roughly 7 for M1, 7 for M2 and 14 for M3 rather than the nominal 27.

Second, the movement lives in the non-US leg, not the US one. On the same samples the non-US leg gives +0.672, +0.383 and +1.309 against the three US-leg zeros above, so the differenced coefficient is essentially the non-US leg with its sign reversed. That is the opposite of the reading the design intends: the triple difference is meant to isolate a US-specific increment, and there is no US-specific movement here to isolate.

Third, the non-US leg is the one contaminated by reporting frequency. Its quarter-on-quarter log change has negative first-order autocorrelation in all 28 countries, median −0.73, which is the semiannual reporting pattern described in Section 1 of the figures note showing up in the outcome. Adding country-by-calendar-quarter fixed effects, the natural correction, cuts the M1 coefficient roughly in half.

We therefore take no position on the differenced block and would rather not build on it. What is needed before it means anything is the portfolio-weight outcome at country level, a rule for the exposure measure's denominator (M1 currently has no floor on the link count: Poland reaches M1 = 0.5 on two recorded links against a sample median of 0.012), and a fix for the reporting sawtooth in the non-US leg. The circular-shift randomization test, the arbiter under our pre-registered hierarchy, has not been run on these cells either.

9. Sample-span variants: a proposal for discussion

The main grid carries every firm-quarter cell, including cells outside the firm's holdings-based existence span. In the audit panel 42.0% of cells sit outside that span. The two estimates sit side by side as follows. Full grid: $\beta_3 = -1.39 \times 10^{-7}$, $p = 0.893$, $N = 909{,}638$, 10,250 firms. Holdings-span restriction: $\beta_3 = -1.84 \times 10^{-7}$, $p = 0.890$, $N = 527{,}380$, 7,988 firms. The null is unchanged; the coefficient moves within a fraction of the standard error.

Our question for you is which sample should be the main specification. Option one promotes the span-restricted sample to main, on the argument that cells outside a firm's observed existence span are not economically at risk of holding changes. Option two keeps the full grid as main and reports the span restriction as robustness, on the argument that the span itself is derived from the holdings data and so conditions on the outcome process. We lean to option two but will follow your call. A third, cleaner variant based on listing coverage windows is not currently buildable: the security coverage file's adjustment-date field is a corporate-actions date, not a delisting date, so a listing-window span needs future symbology data.

10. Extensive versus intensive variation in the three measures

Method, in brief. For each country and adjacent quarter pair we restrict to firms present in both quarters, split them into firms that switch any-China-link status (extensive) and firms linked in both quarters (intensive), and compute a two-player Shapley decomposition of the change in each measure; the gap between this balanced-set change and the actual measure change (panel entry and exit, plus denominator and weight drift among never-linked firms) is a residual, and variance shares are covariance betas of each component on the actual change across country-quarter cells.

The intensive margin contributes little under M1 and M3, at 5 and 4 percent, but carries 35 percent under M2. The extensive and composition margins carry most of the variation, and which of the two leads depends on the measure. For M1 the shares are 57% extensive, 5% intensive, 39% residual (composition). For M2 they are 51% extensive, 35% intensive, 14% residual. For M3 they are 35% extensive, 4% intensive, 61% residual; the equal-weighted measure is the most exposed to universe composition because every entrant shifts the mean directly. Within the balanced continuing-firm set the extensive share rises to 93% (M1), 60% (M2), and 91% (M3). The quarter-to-quarter movement in country exposure is mostly firms gaining or losing China links, and universe composition, not deepening of link counts among firms already linked. M2 is the partial exception: cap-weighting gives continuing large firms a substantial intensive share.

11. The value-weighted measure is conditional on coverage

Market cap covers a declining share of the exposure-defined universe. In the estimation universe the covered share of firm-quarters is 66.6% in 2010, 48.0% in 2015, 45.5% in 2020, and 42.9% in 2023. At the distinct-firm level the 2023 figure is 46.9%. M2 is therefore an average over the covered subset only, which tilts toward larger listed firms, and its coverage falls over time as the Revere-matched universe adds small unlisted firms. We report the covered share for every country-quarter cell and read M2 jointly with M1 and M3 rather than on its own. In the full Revere universe M2 is not computable at all, since unmatched firms have no market cap in the data.

12. Figures A and B: status and read

The figure package agreed on 4 August is built on the v3.1 panel and current as of 21 August. Figure A exists at the firm level in a rolling zero-tie-versus-positive-tie version, a chain-linked matched-sample version that removes roster changes by construction, and equal-weighted and market-cap-weighted quartile versions; at the country level it exists in one version per exposure measure (M1, M2, M3), as requested. Figure B exists in a two-bucket (no-US versus any-US) and a three-bucket (no-US, low-US, high-US) version. All nine figures carry the tension index as background. The full construction notes are in the captions of the accompanying note.

The read on Figure A. Two denominators are available and they are an order of magnitude apart, so the label matters. Against the US investors' full global equity book, which is the denominator the regressions use and the one the figures plot, the zero-tie share falls from 5.15 percent in 2018Q4 to 3.64 percent in 2022Q4 while the positive-tie share is 7.53 and 7.55 percent. Against the European book alone, which is the within-Europe reallocation diagnostic, the same three cells read 37.2 percent, 29.6 percent and 61.3 percent. The chained matched-sample version is far flatter than the rolling one: the two arms end the sample within ten points of each other, so a large part of the raw wedge is composition (the data reclassifying firms into the positive arm) rather than reallocation. Any discussion should lead with the chained panel.

The read on Figure B runs opposite to the hypothesis it was built to test. Over 2018 to 2023, year-over-year growth in China links averages 19.1 percent for firms with US institutional owners against 9.3 percent for firms with none, with ranges of 3.1 to 37.3 percent and 0.6 to 20.0 percent respectively. The ordering is not uniform: the no-US group grows faster in 6 of those 24 quarters. No pre-emptive de-linking is visible in any quarter.

13. Proposed ordering of next steps

1. Figures discussion and sign-off (the package above), including the composition caveat and whether the chained matched-sample framing becomes the lead.

2. The sample-span decision from Section 9, which gates the full battery rerun.

3. Investor heterogeneity, in two stages: first the state-dependence classification (tax-exempt endowments, charities, pension funds are the cheapest to construct), then the mutual-fund versus separate-account comparison as a falsification test.

4. Firm-side response beyond the descriptive Figure B.

5. Extensions (reverse direction, other countries' investors, other geopolitical pairs) only after the single-direction result is settled.